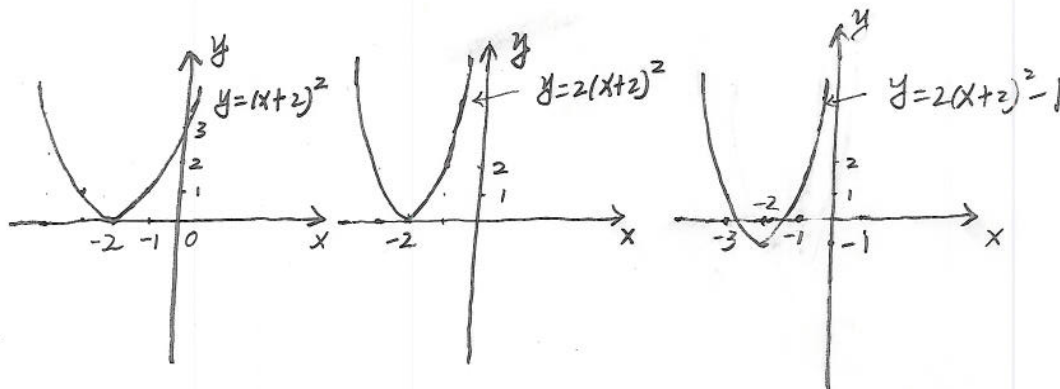


Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (4 points) Sketch the graph $y = 2(x+2)^2 - 1$ by making appropriate transformations to the graph of $y = x^2$.



Problem 2. (6 points) Solve for x without using a calculating utility.

(a) $\log_2(2^{3x}) = 9$.

Solution:

$$\begin{aligned}\log_2(2^{3x}) &= 9 \\ (3x)\log_2 2 &= 9 \\ 3x &= 9 \\ x &= 3\end{aligned}$$

(b) $e^{-2x} - e^{-x} - 6 = 0$.

Solution: Taking $u = e^{-x}$, then we have.

$$(e^{-x})^2 - e^{-x} - 6 = 0$$

$$u^2 - u - 6 = 0$$

$$(u-3)(u+2) = 0$$

$$u = 3 \text{ or } u = -2$$

i) $u = 3 \Rightarrow e^{-x} = 3 \Rightarrow -x = \ln 3 \Rightarrow x = -\ln 3$

ii) Since $e^{-x} > 0$, there is no x satisfying $e^{-x} = -2$.

So this equation only have one solution $x = -\ln 3$